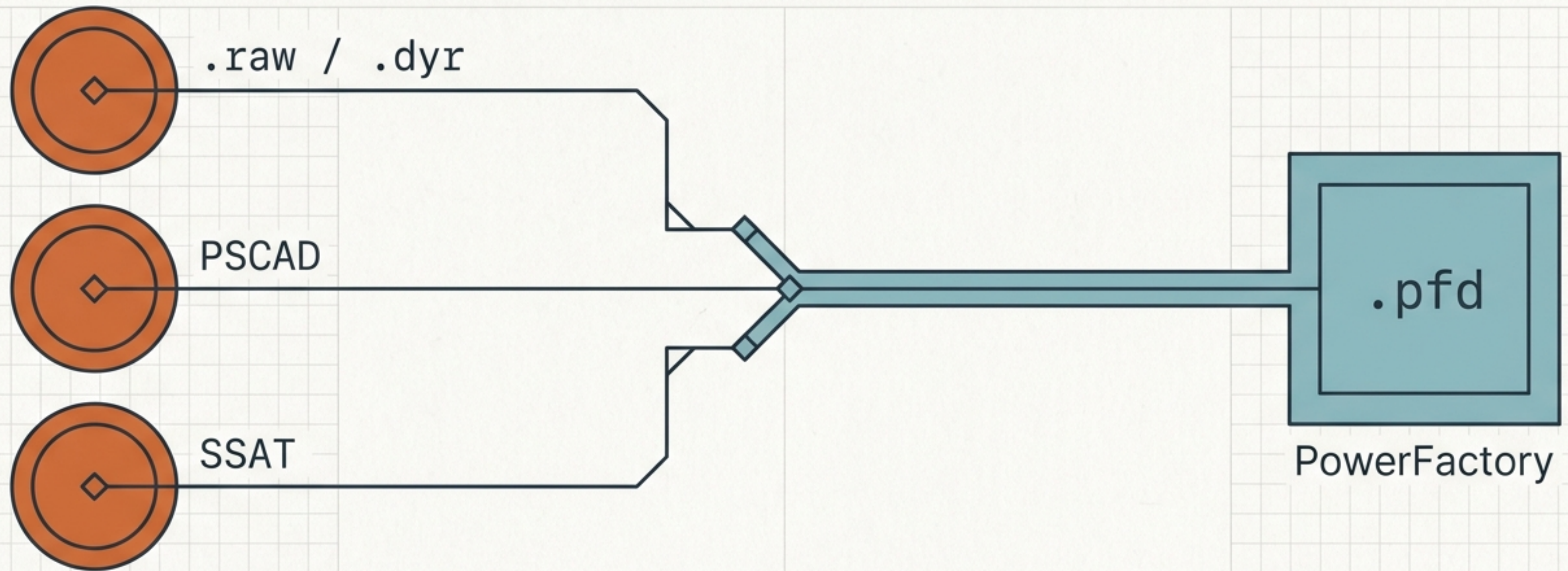
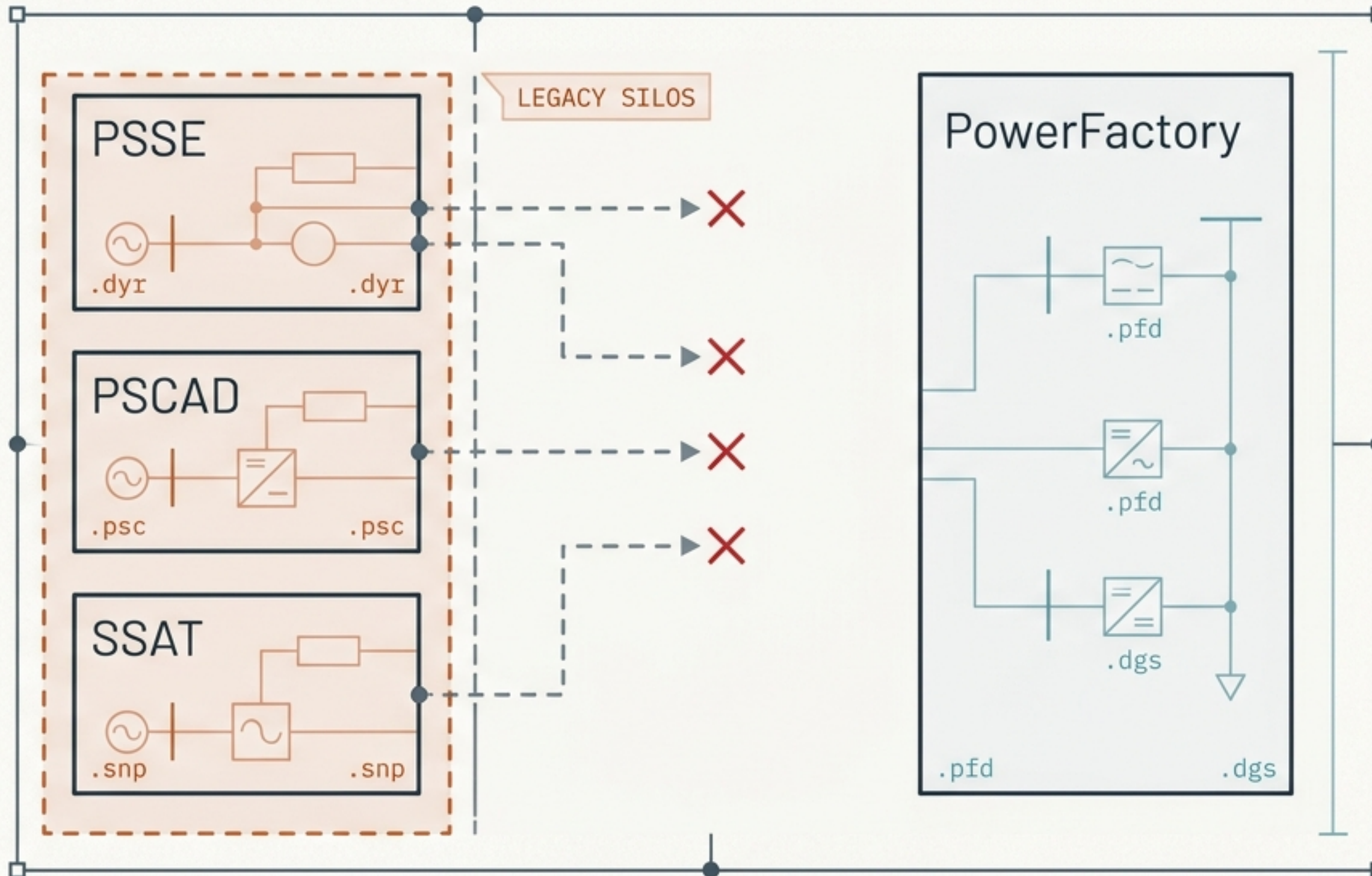


Dynxlate: The Rosetta Stone for Dynamic Model Translation

Bridging PSSE, PSCAD, SSAT, and PowerFactory.

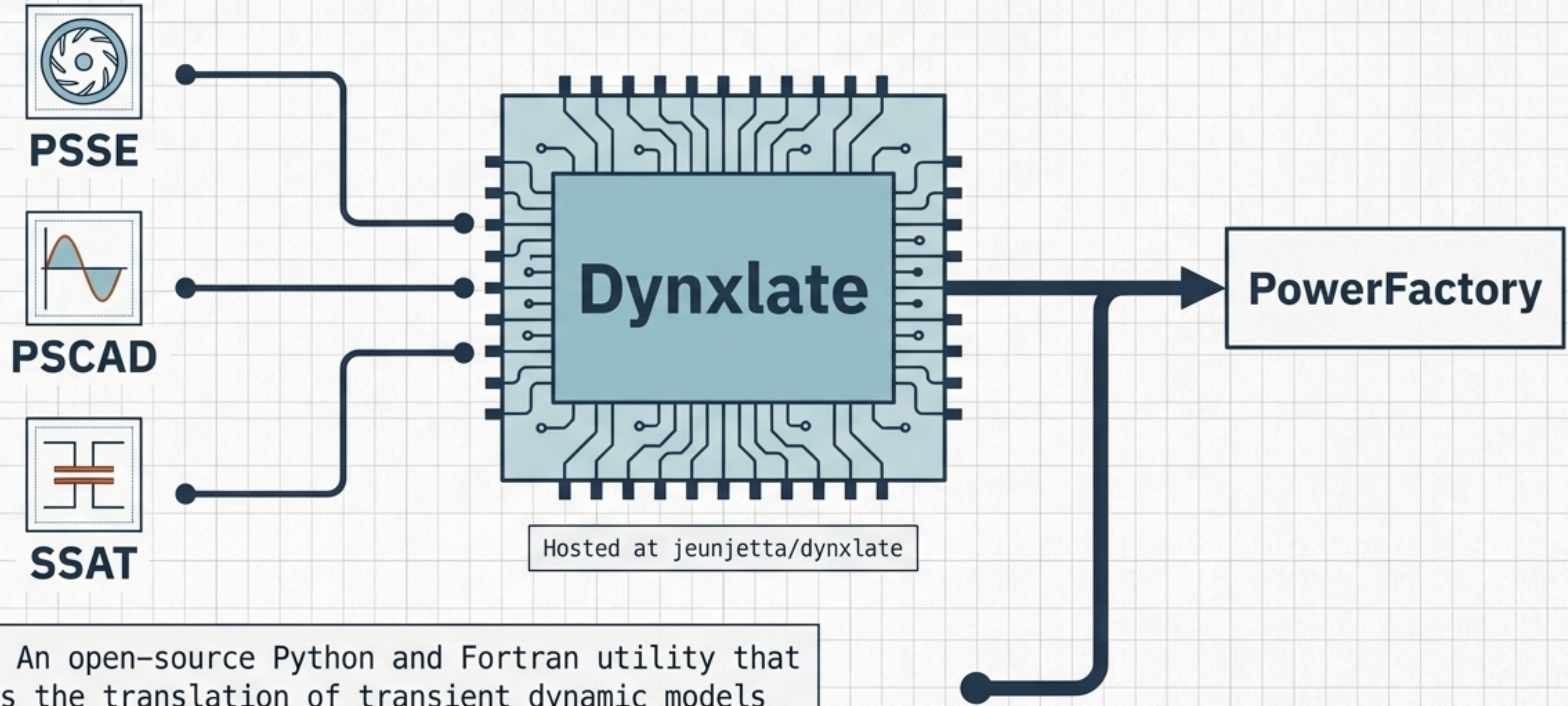


The Language Barrier in Grid Modeling



Transient dynamic models (generators, SVCs, inverters) are locked in proprietary formats. Manual recreation introduces transcription errors, wastes engineering hours, and prevents seamless grid interoperability.

Automating Cross-Platform Interoperability

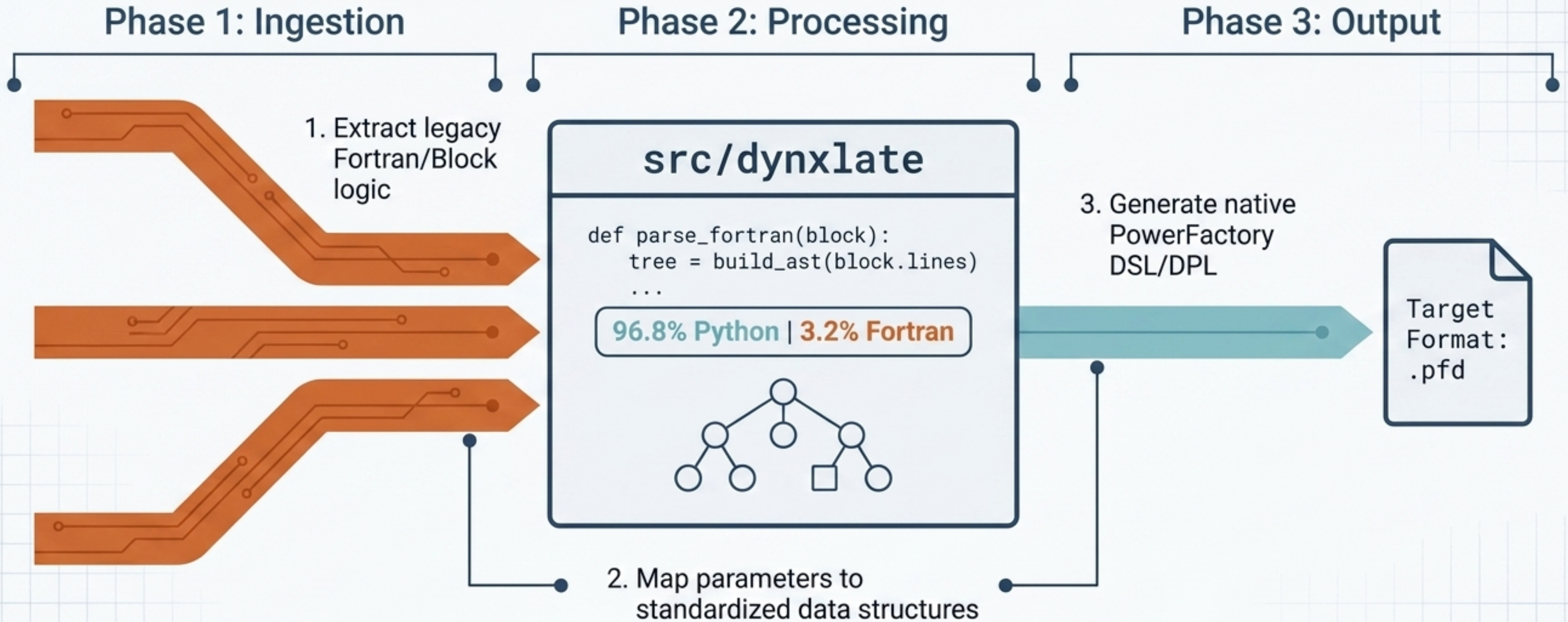


Thesis: An open-source Python and Fortran utility that automates the translation of transient dynamic models across industry-standard simulation software.

The Simulation Ecosystem Matrix

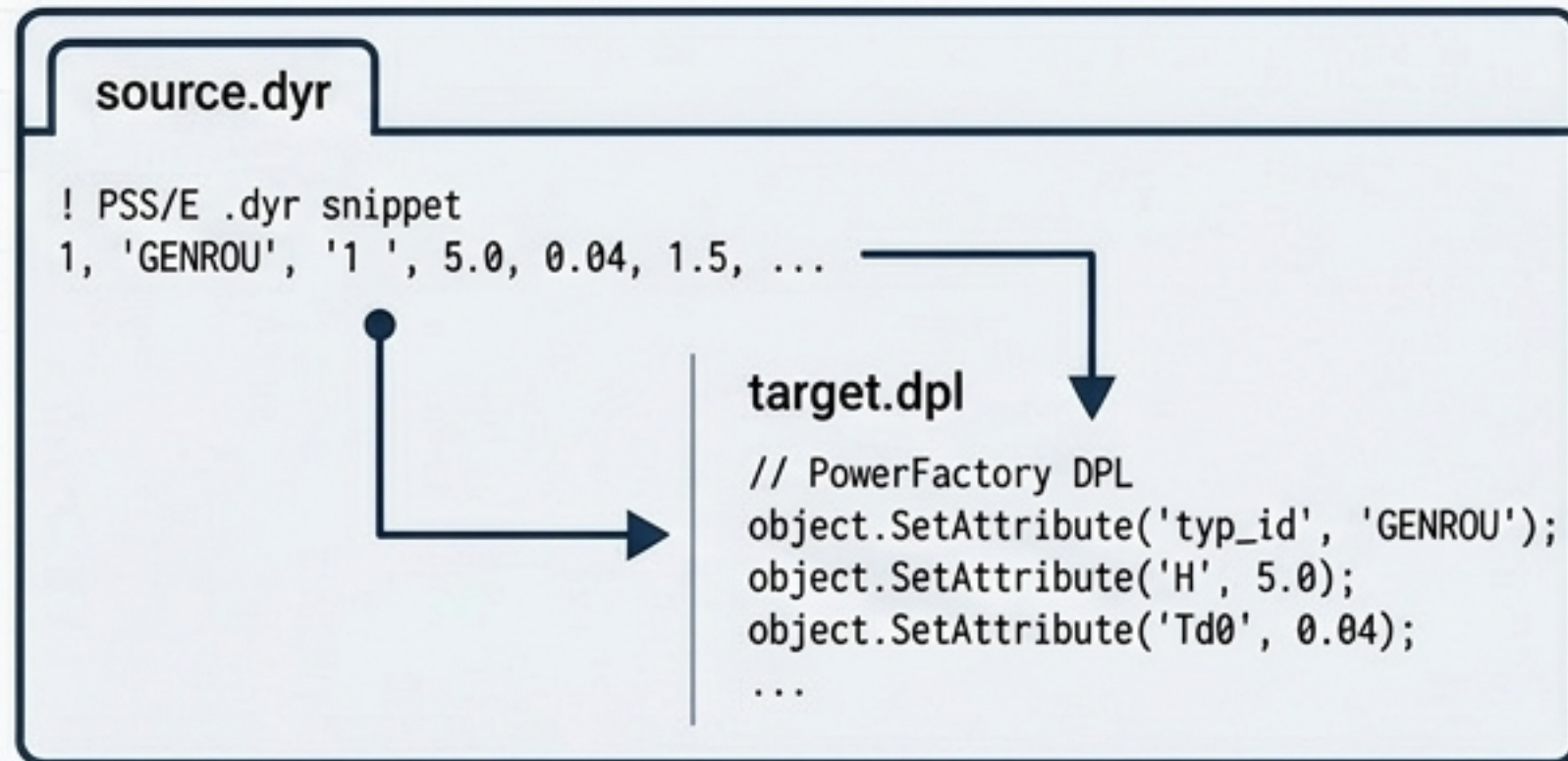
	PSSE	PSCAD	SSAT	PowerFactory
Role in Dynxlate	Source	Source	Source	Target
Native File Formats	.raw + .dyr	Case Files	SSAT formats	.pfd
Model Paradigms	Block Diagram & Fortran	Graphical Block Diagram	Analytical	DPL / DSL

The Translation Engine Map



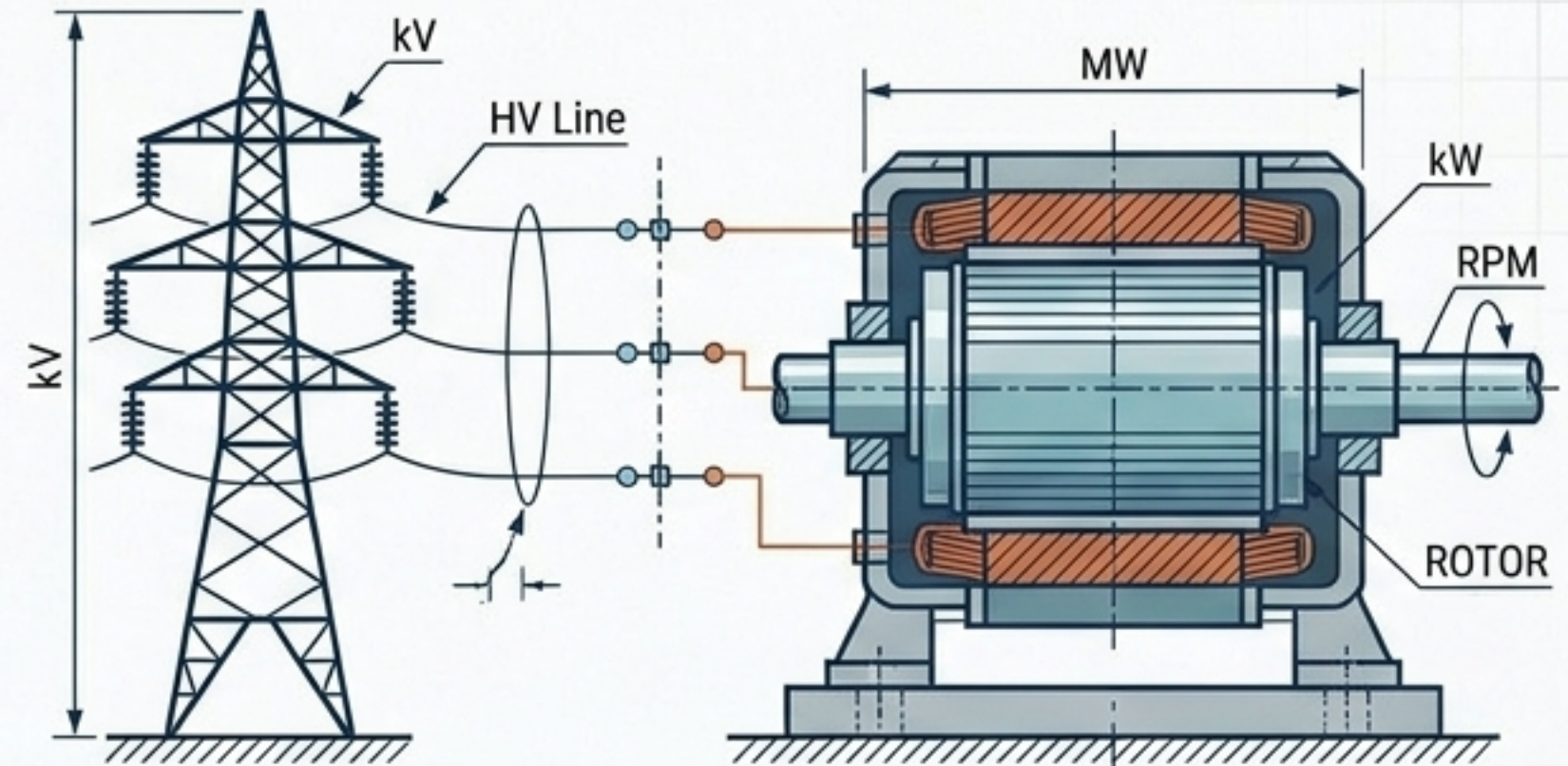
Code Translation Requires Physical Verification

Syntax Parity



Syntax translation ensures the software compiles...

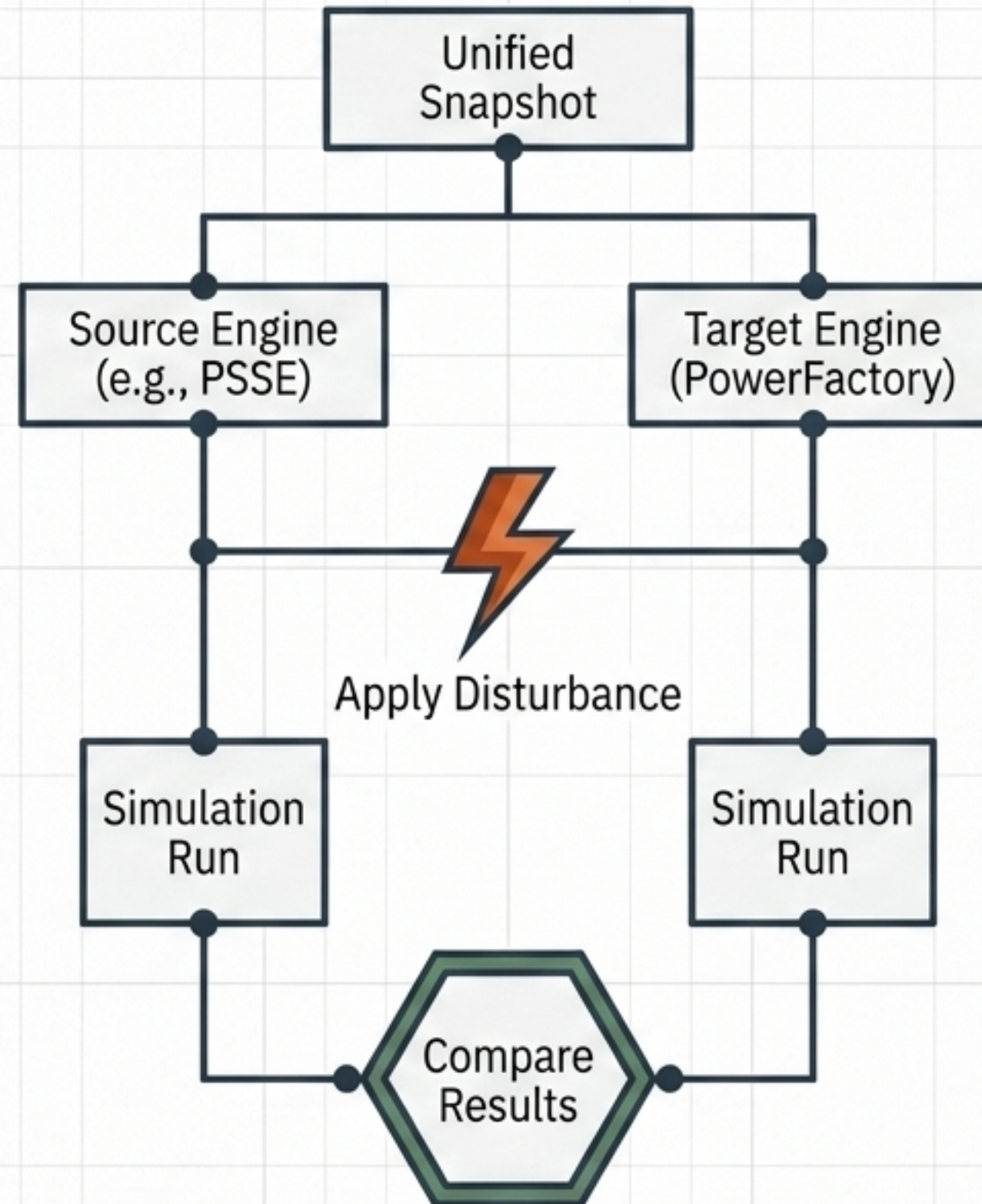
Physical Parity



...but physical verification guarantees the grid behaves identically.

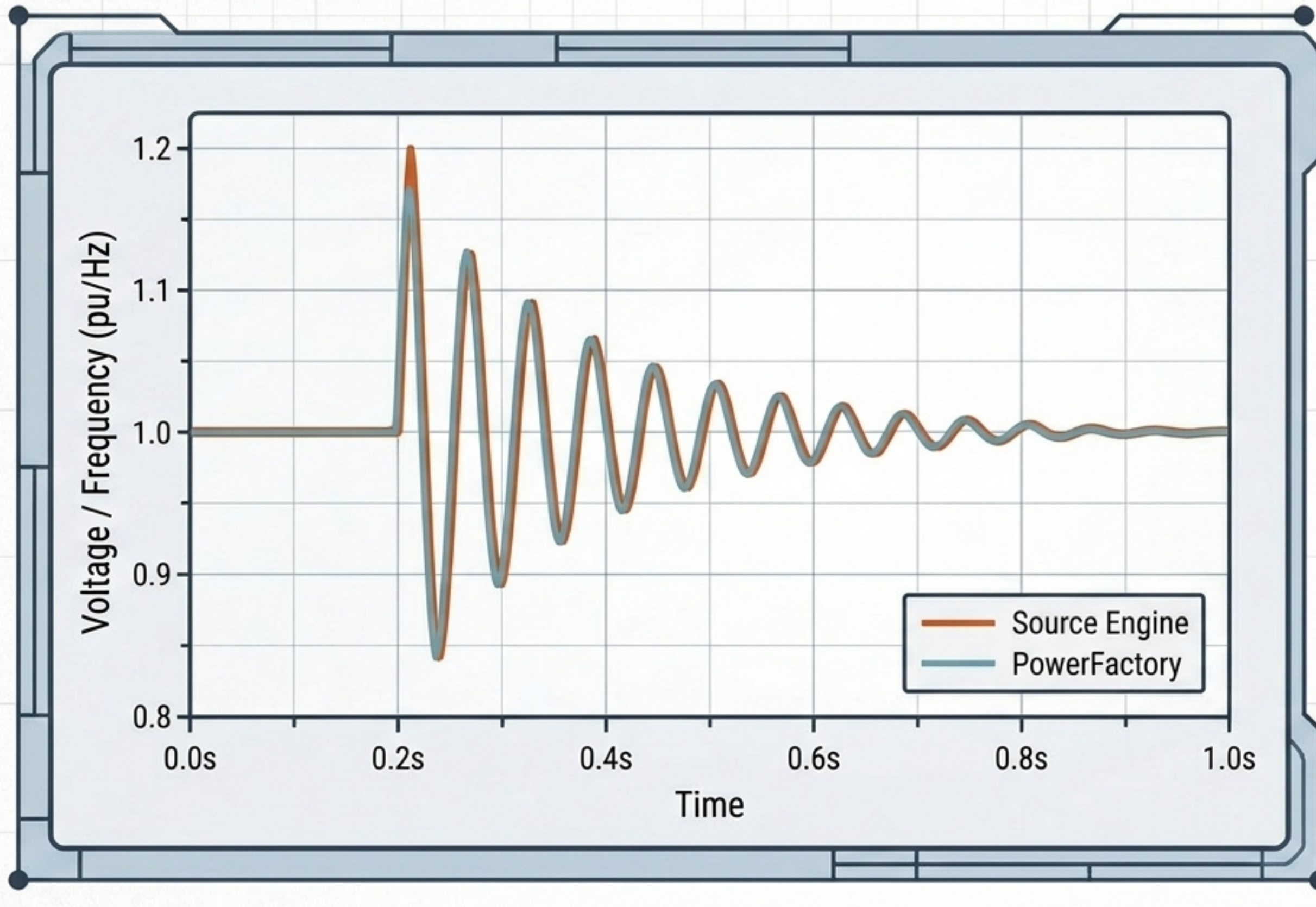
Translating .raw to .pfd means nothing unless transient behaviors match perfectly under stress. Dynxlate integrates verification directly into its core workflow.

The Verification Dual-Trace Loop



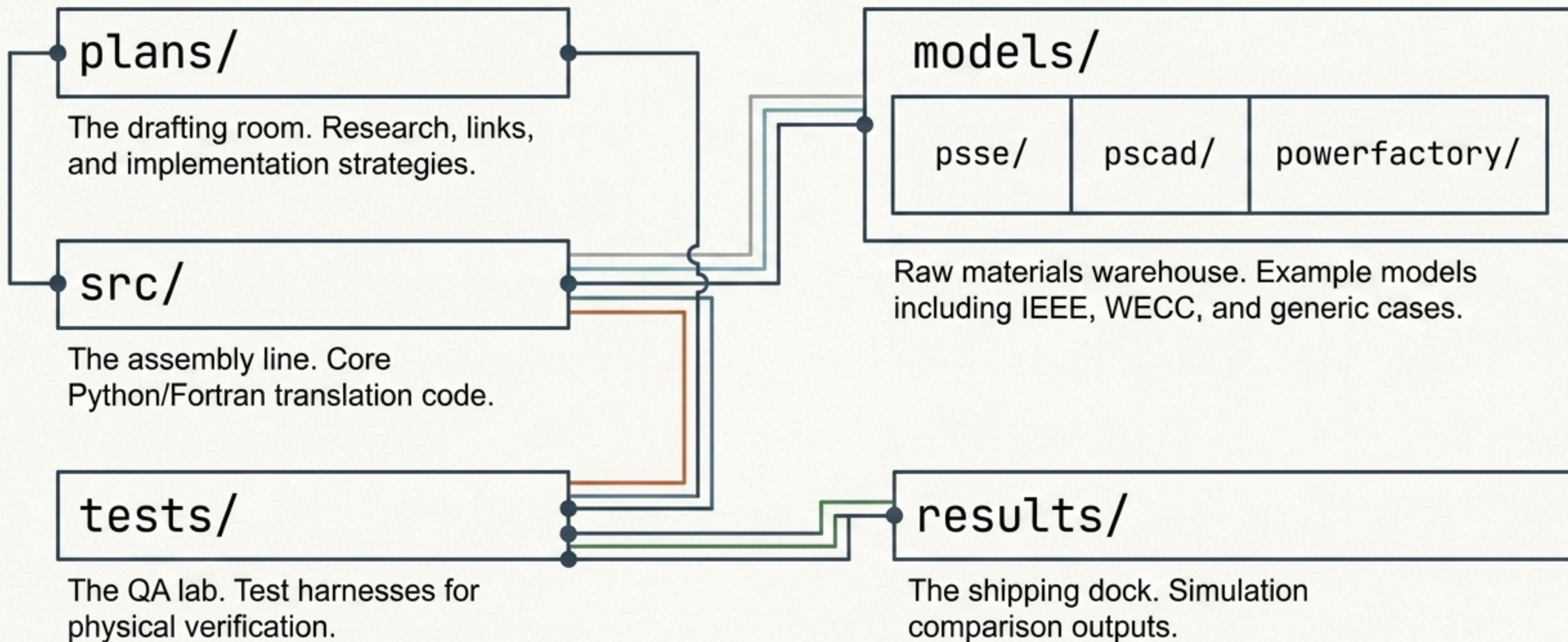
Dynxlate validates its own translations by running the exact same network snapshot and disturbance through both engines simultaneously.

Guaranteeing Transient Dynamic Parity



Match Verified:
Mathematical parity
achieved across
platforms.

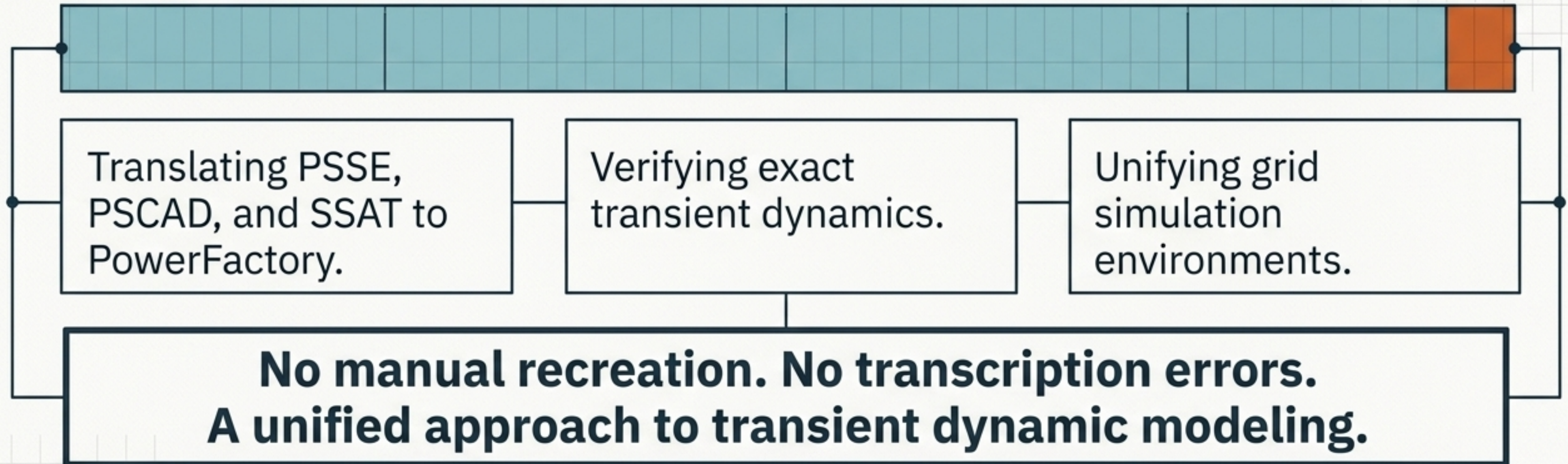
Repository Topography



Interoperability Built on Open Source

96.8% Python

3.2% Fortran



```
$ git clone https://github.com/jeunjetta/dynxlate
```